Singrauli Coalfield

Singrauli is one of India's most important industrial regions, widely known as the *"Energy Capital of India."* Located at the border of Madhya Pradesh and Uttar Pradesh, it plays a crucial role in meeting the country's growing energy demands. The region is rich in coal reserves, making it a major hub for large-scale mining and power generation activities.

The development of Singrauli accelerated after the discovery of vast coal deposits in the mid-20th century. Today, it is home to several major mining and power companies such as Northern Coalfields Limited and NTPC Limited, which contribute significantly to India's electricity production.

Apart from its industrial importance, Singrauli represents a complex balance between economic development and environmental sustainability. While it has boosted employment and infrastructure, it has also raised concerns related to pollution, deforestation, and displacement of local communities.

GEOGRAPHICAL OVERVIEW

Singrauli is situated in the northeastern part of Madhya Pradesh and extends into the southeastern region of Uttar Pradesh. The region mainly falls within the Singrauli district of Madhya Pradesh and Sonbhadra district of Uttar Pradesh.

📍 Location & Boundaries

- Lies between the Vindhyan ranges
- Close to the border of multiple states, enhancing connectivity
- Strategically located for industrial development

🌅 Physical Features

- Part of the **Vindhyan Plateau**, characterized by rocky terrain
- Rich in mineral resources, especially coal
- Forested areas (though significantly reduced due to mining)

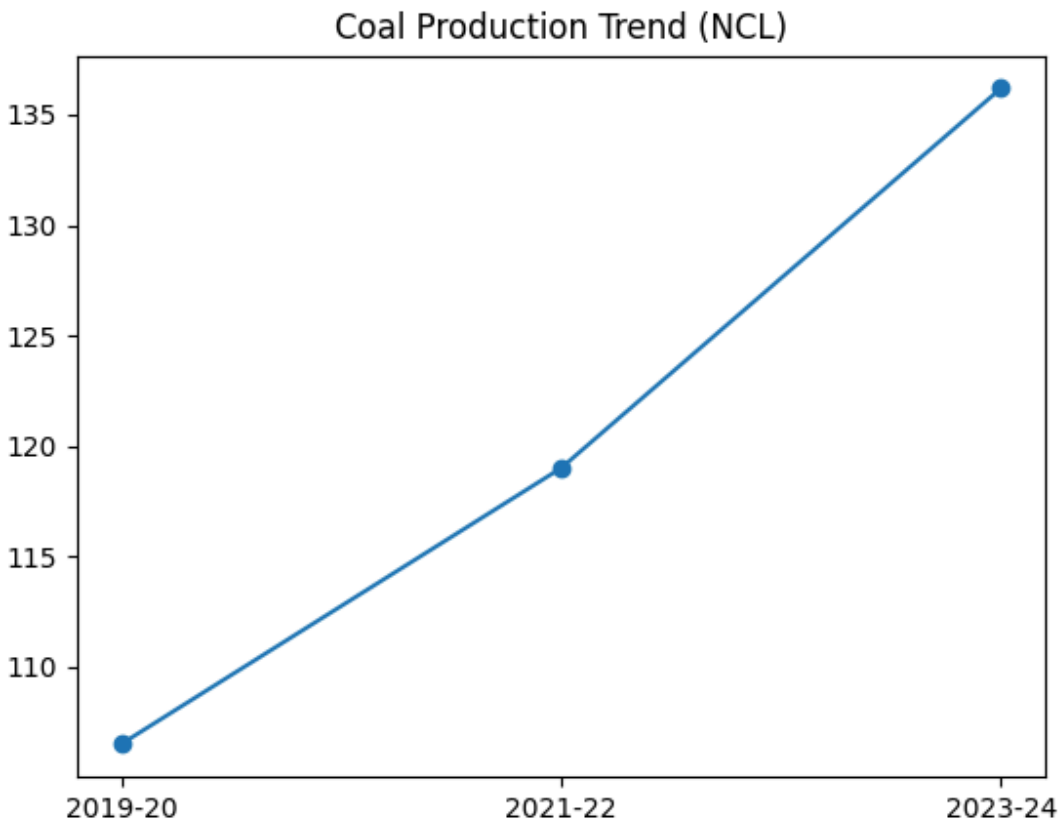
🌦️ Climate

- Tropical climate
- Hot summers (temperatures often above 40°C)
- Moderate monsoon rainfall
- Cool winters

🌱 Natural Resources

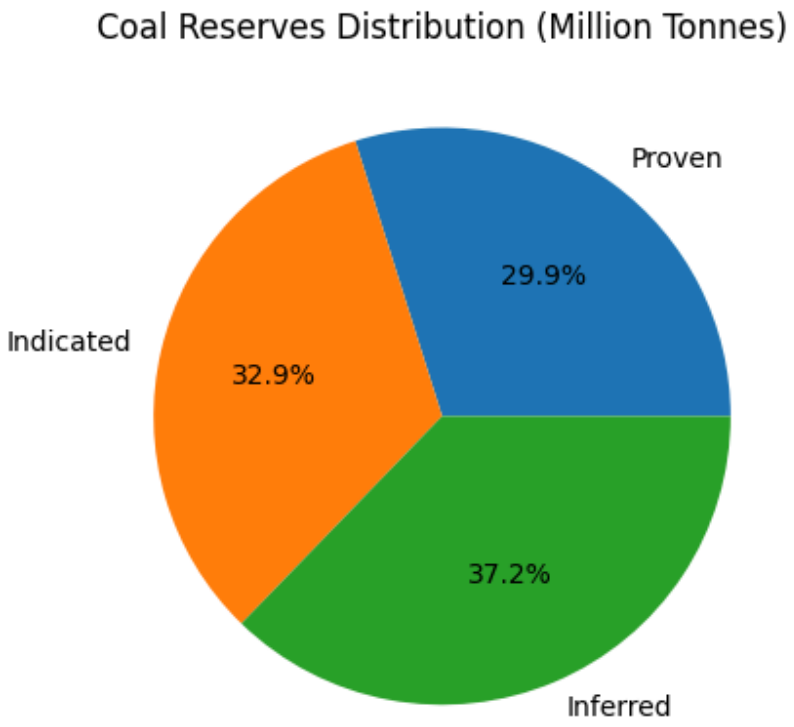
- Vast coal reserves (one of the largest coalfields in India)
- Availability of water and land made it ideal for power plants and mining operations

📊 1. Coal Production Trend (NCL)



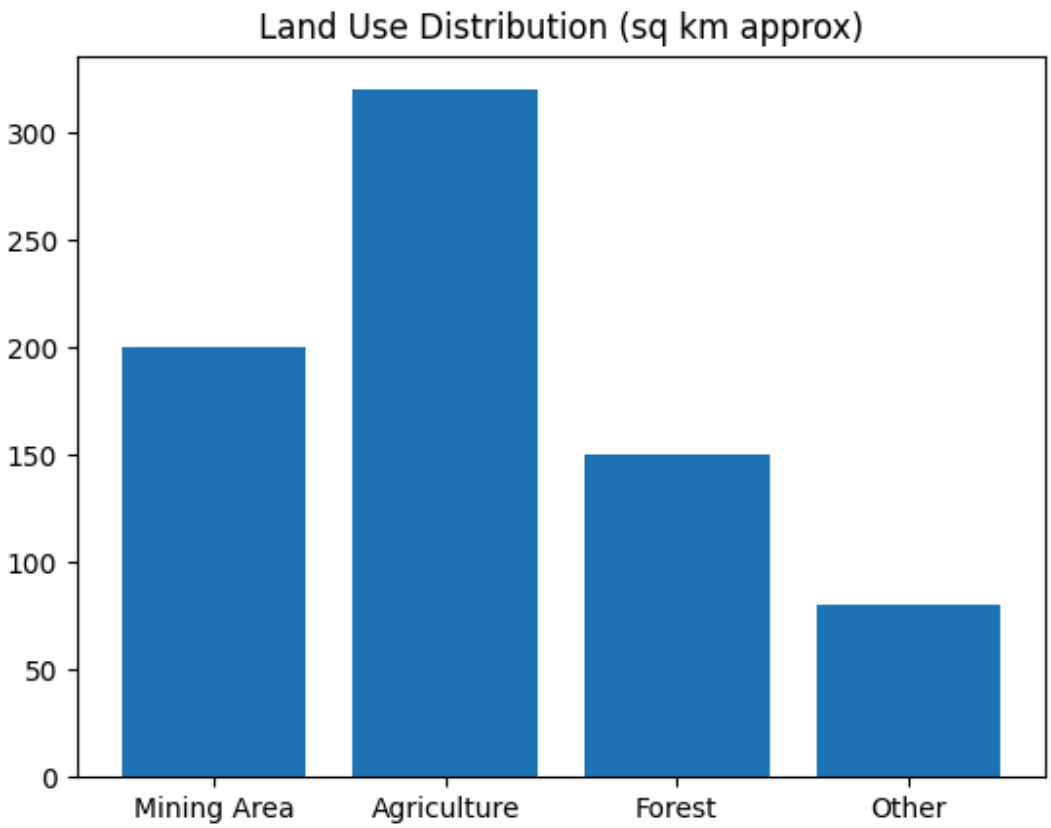
This graph shows the steady increase in coal production over recent years in Singrauli, mainly driven by Northern Coalfields Limited.
It reflects improved mining efficiency, mechanization, and rising energy demand.
Significance: Helps understand production planning, capacity expansion, and performance benchmarking.

📊 2. Coal Reserves Distribution



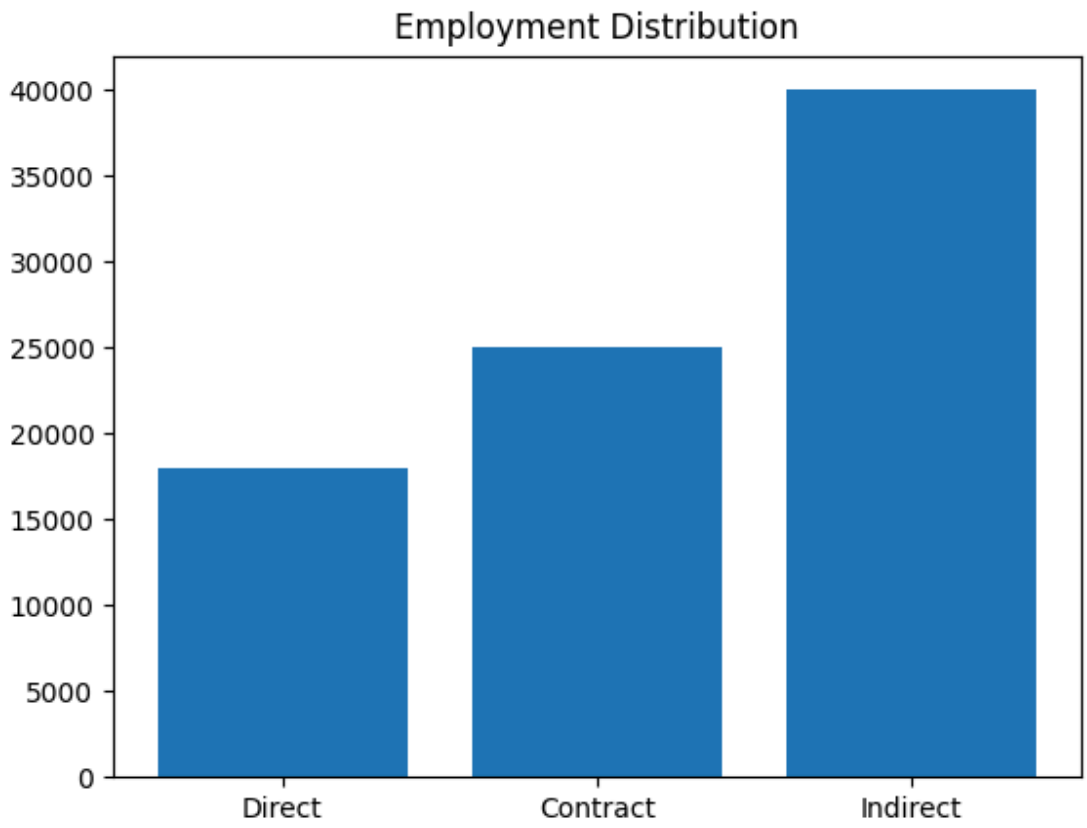
This chart divides coal reserves into proven, indicated, and inferred categories.
It shows how much coal is currently extractable versus what is still under exploration.
Significance: Crucial for mine planning, feasibility studies, and long-term resource management.

📊 3. Land Use Distribution



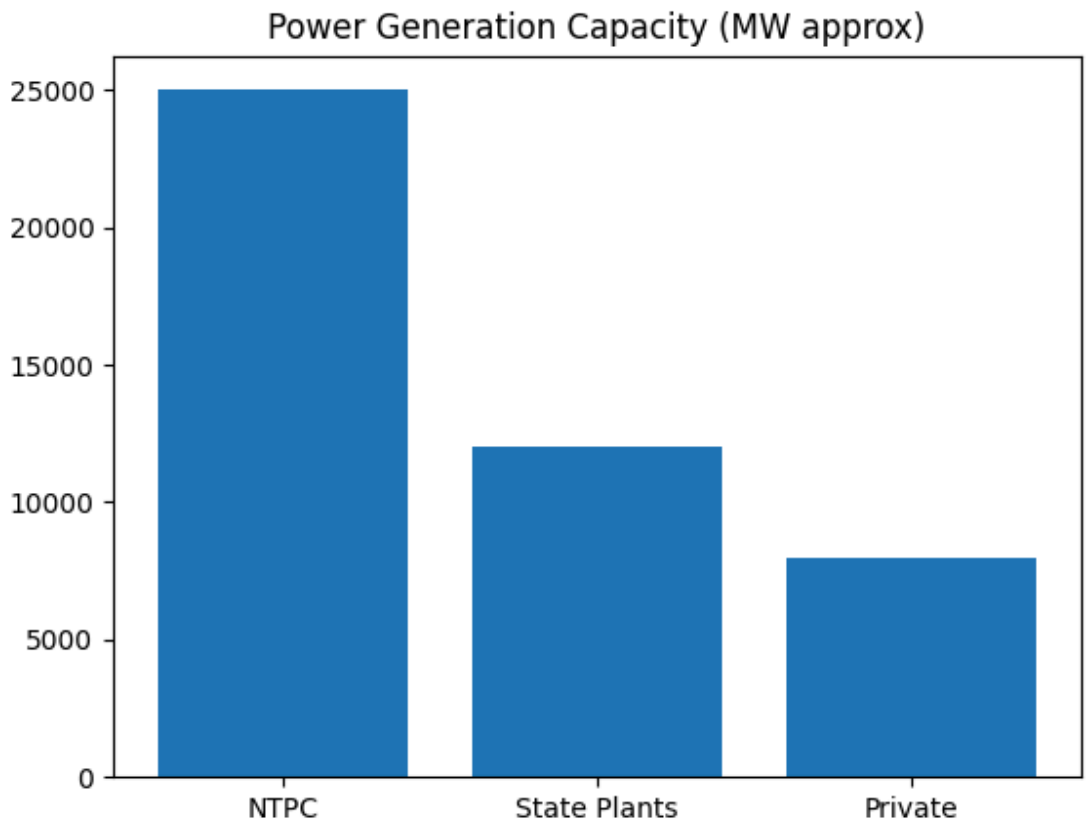
This visualization highlights how land is divided among mining, agriculture, forest, and other uses.
Mining occupies a significant portion, often replacing natural land cover.
Significance: Important for environmental impact assessment (EIA) and land reclamation planning.

📊 4. Employment Structure



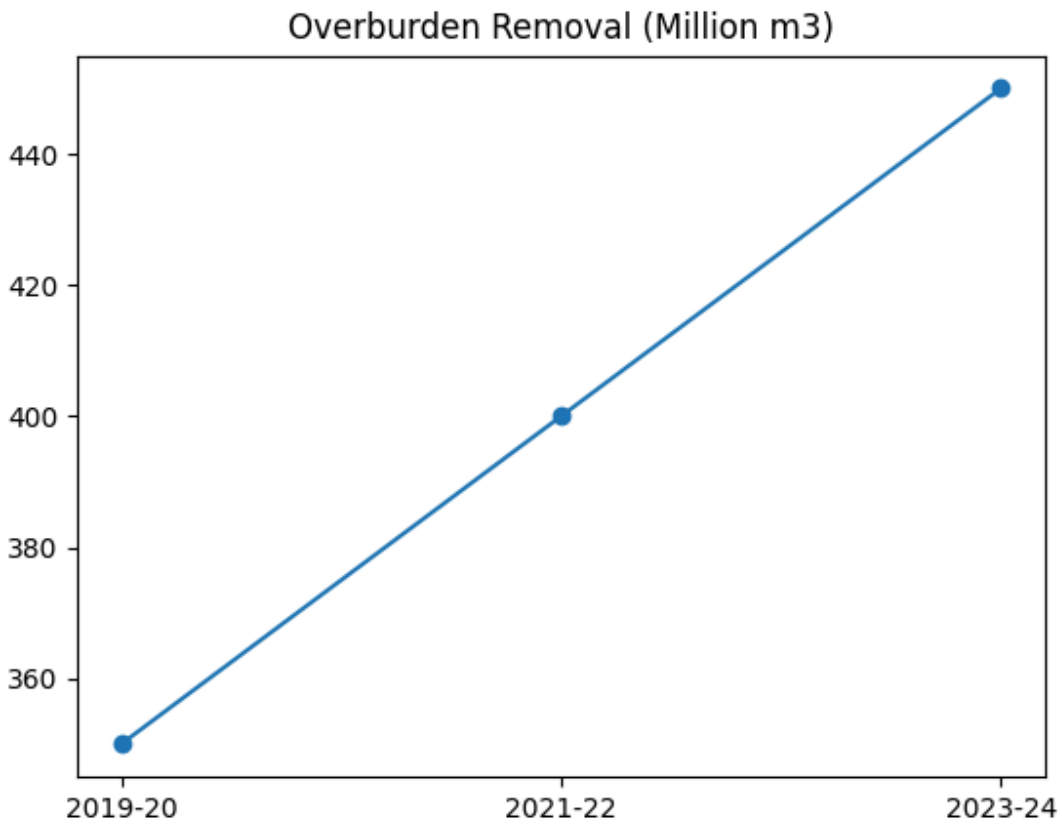
This chart shows the distribution of workforce into direct, contract, and indirect employment. Mining generates large-scale livelihood opportunities beyond just core operations.
Significance: Useful for understanding labor economics and workforce management in mining projects.

📊 5. Power Generation Capacity



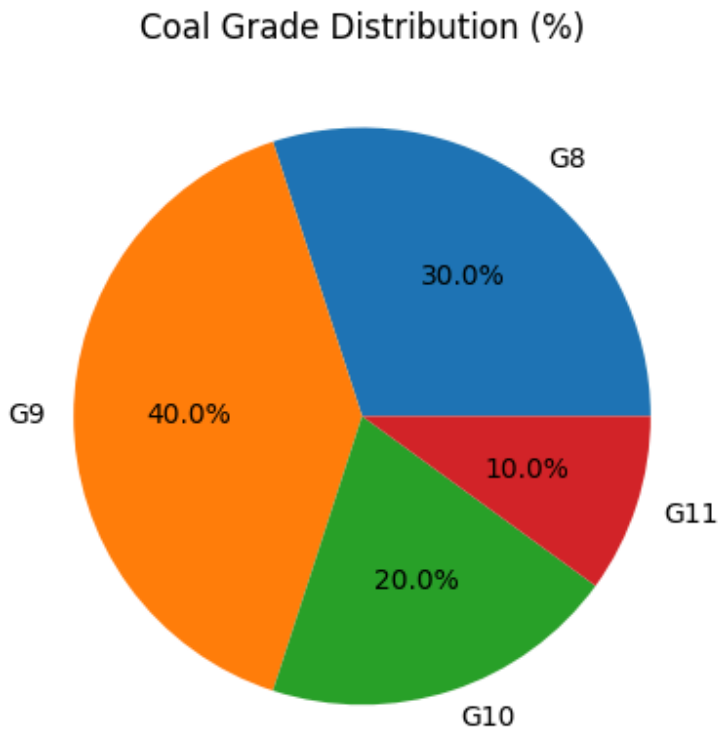
This graph represents how coal from Singrauli supports thermal power generation by companies like NTPC Limited.
It highlights the region's contribution to electricity supply.
Significance: Demonstrates the direct link between mining output and national energy security.

📊 6. Overburden Removal Trend



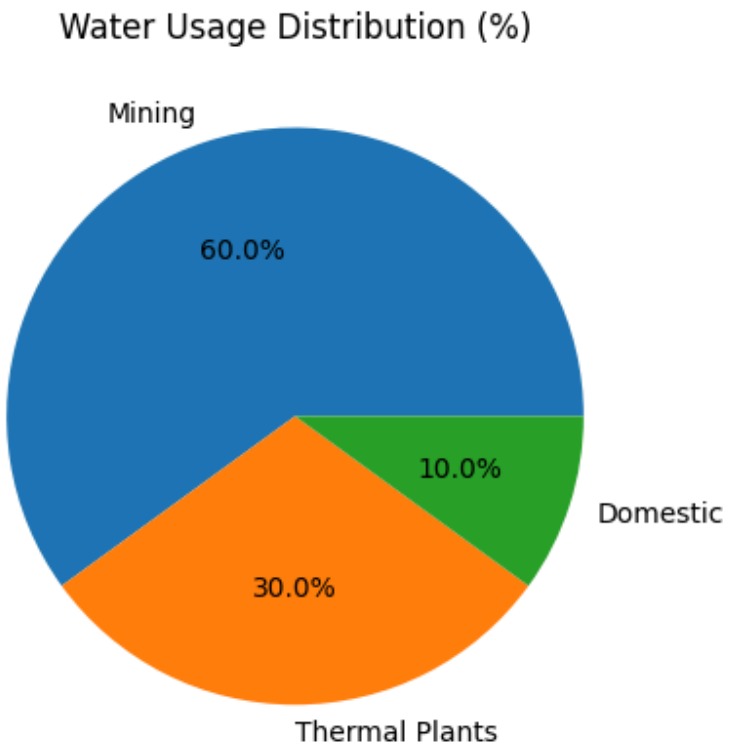
This chart shows the amount of overburden (waste material) removed to access coal seams. It increases with deeper mining and expansion of pits.
Significance: A key mining parameter—used to calculate stripping ratio, cost, and mine efficiency.

📊 7. Coal Grade Distribution



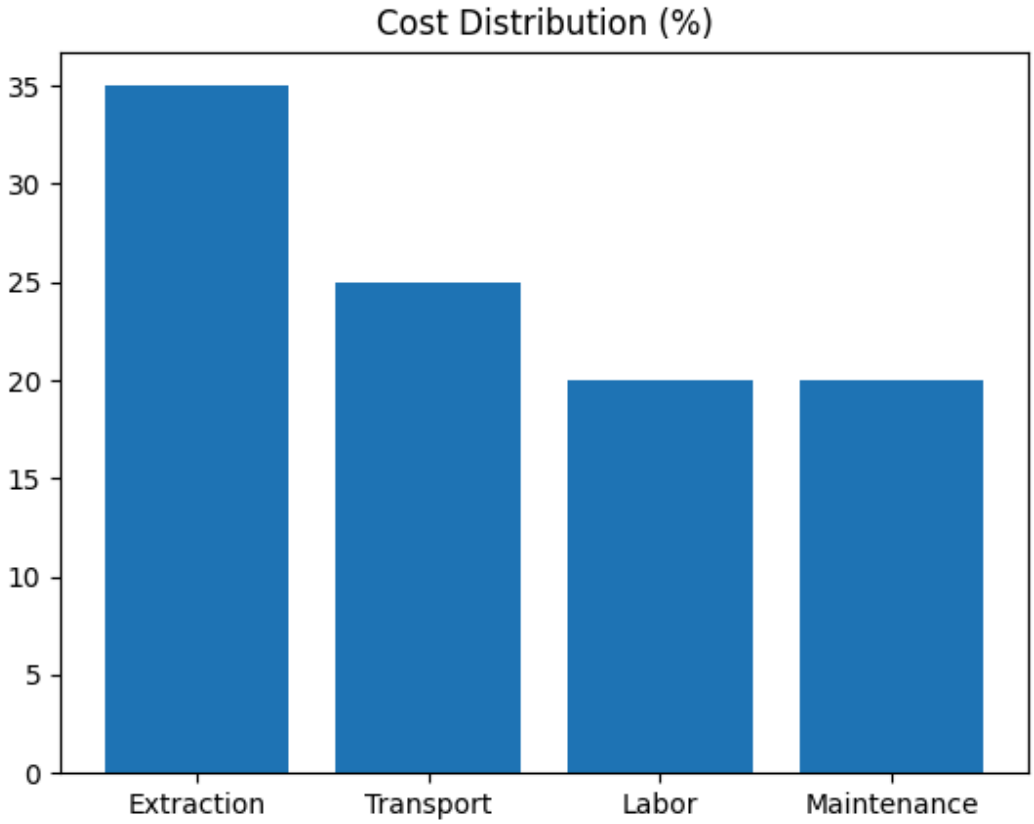
This chart shows the proportion of different coal grades (G8–G11) found in Singrauli.
Different grades indicate varying calorific values and quality.
Significance: Essential for deciding coal usage, pricing, and suitability for thermal plants.

📊 8. Water Usage Distribution



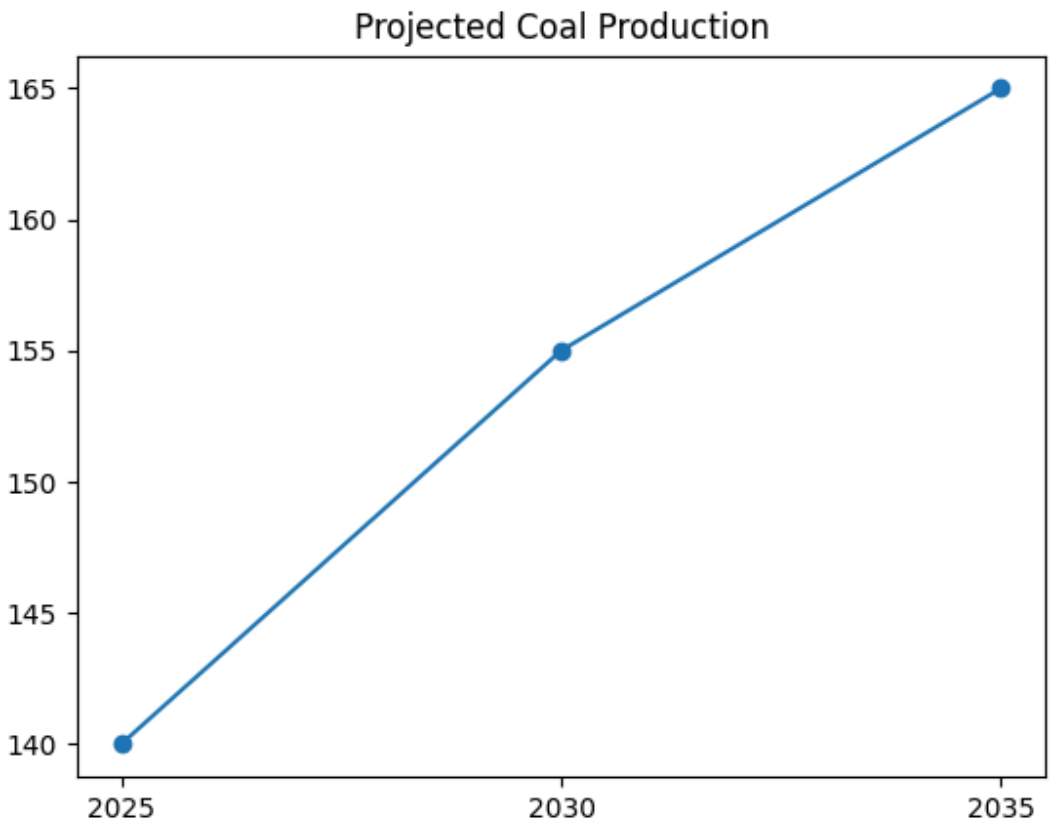
This chart represents how water is consumed across mining, power plants, and domestic use. Mining and thermal plants require large volumes of water.
Significance: Important for water resource management and sustainability planning.

📊 9. Cost Distribution in Mining



This graph breaks down major cost components like extraction, transport, labor, and maintenance.
Transport and extraction typically form the largest share.
Significance: Helps in cost optimization, budgeting, and improving operational efficiency.

📊 10. Future Production Projection



This chart predicts future coal production based on current trends and expansion plans. It reflects increasing demand for energy and planned capacity growth.
Significance: Useful for strategic planning, policy-making, and investment decisions.